



Development and Evaluation of a Maternal Health Risk Classification Algorithm on Internet of Things-Derived Physiological Data

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Abstract: Maternal health forms a pivotal public health concern since poor pregnancy outcomes often arise from delayed detection of physiological abnormalities. Recent advances in Internet of Things (IoT)-based maternal health monitoring devices have enabled automated risk assessment of real-time clinical data. This study developed a Machine Learning (ML) framework to categorize maternal health risk into High, Medium, and Low Levels using a dataset of 1,014 records collected from maternal health centres. Six physiological indices, including body temperature, heart rate, systolic, and diastolic blood pressure, blood glucose level, and age were analyzed via five-fold cross validation. Logistic Regression (LR), Decision Tree (DT), and Random Forest (RF) models were trained and evaluated. RF model achieved the best performance of 0.88 accuracy and F1-score of 0.88. Taken together, recall of 0.91 for high-risk cases, area under the curve (AUC) of 0.973, log loss of 0.449, mean squared error (MSE) of 0.228, root mean squared error (RMSE) of 0.478, and mean absolute error (MAE) of 0.157 demonstrated reliable prenatal risk prediction despite limited resources in healthcare settings.

Keywords: Maternal health; IoT-based monitoring; Machine learning; Ensemble; Physiological parameters; Low-resource

1. Introduction

Maternal health remains a major public concern, particularly in low- and middle-countries, where inadequate access to quality maternal healthcare and delayed identification of pregnancy-related complications can contribute to adverse maternal outcomes. Medical specialists and researchers are finding novel approaches for improved maternal health outcomes arising from pregnancy-associated mortality. Diabetes, obesity, maternal depression anxiety, and elevated blood pressure are common pregnancy-related problems. According to the World Health Organization (WHO), a woman passes away every two minutes from excessive blood pressure or any other pregnancy-related issues (World Health Organization, 2023).

Significant efforts are directed at duties like patient care, diagnosis, therapy, and other procedures when Machine Learning (ML) algorithms are integrated into the healthcare industry. Predictive analytics for different diseases, disease identification, automation, development of preventative, curative programs, and patient monitoring systems are just a few applications where ML algorithms could produce results. Besides, solutions like risk assessments, appropriate resource allocation, drug discovery and development, and treatment regimens could be aided by the application of ML algorithm (Khadidos et al., 2024). The dataset, originally curated by Ahmed & Kashem (2020) from Internet of Things (IoT) monitored maternal health centres in Bangladesh, comprised 1,014 records with six maternal physiological features including body temperature, systolic blood pressure, heart rate, blood glucose level, diastolic blood pressure, and age, to be classed as low, medium, and high risk. This dataset has later been adopted across several studies as the standard by which maternal risk classification algorithms are

judged.

Maternal health is so vital that researchers have created models and methods for timely prediction of health concern during pregnancy using both traditional and ML methods. While some research aimed at prediction, others examined states of pregnant women in order to identify and document health risk factors. For example, a Prediction Risk Detection System (PRDS) was designed to determine the risk level of pregnancy depending on the symptoms the expectant mother has experienced (Krisnanik et al., 2019). Also, telemedicine interventions powered by AI-driven virtual assistants could improve outcome and bridge access gaps to health care in underserved and rural communities; this potentially improve results and lessen access inequalities. In brief, ML may serve dual purpose towards improving clinical care and fostering greater fairness in maternal health services, especially in low-resource setting (UNICEF, 2024). According to Tzimourta et al. (2025), healthcare professionals may proactively use ML techniques in maternity care by concentrating treatment on areas and situations where they are most needed. Mothers and babies will benefit from this in the short and long term.

Previous works focused on evaluation practices suitable for low-resource maternal health scenes, where there is deployment constraint, limited computational infrastructure, sparse clinical staff for model oversight, and cost of false negative on high-risk population (Ahmed & Kashem, 2020; Khadidos et al., 2024; Mashrafi et al., 2024). These studies contributed not just to the re-demonstration of classifier's accuracy on familiar benchmark, but also further evaluated models using broader set of complementary metrics: accuracy, area under the curve (AUC), log loss, and error-based measures. Attention was particularly drawn on recalling high-risk class, where missed detections carry the greatest clinical cost. They showed that a lightweight and interpretable classifier could match or exceed heavier ensemble methods of this task; this is a result relevant to settings where computational resources and technical capacity for model maintenance are limited.

2. Literature Review

Maternal health refers to the general state of a woman's physical, emotional, and mental well-being during pregnancy, childbirth, and the postpartum phase. Monitoring maternal morbidity and mortality rates is crucial because they prove the availability and accessibility of healthcare services. Maternal mortality is chiefly caused by a few pregnancy-related complexities; these often include diabetes, hypertension, preterm birth, and excessive bleeding. Premature birth and maternal death were among the serious issues that could be avoided with prompt identifying and addressing pregnancy-related risks (Mashrafi et al., 2024). Globally speaking, health complications and deaths related to pregnancy, childbirth, and postnatal periods remain serious health issues. Consequently, reducing the maternal mortality ratio (MMR) is a crucial metric in the Millennium Development Goals (MDGs), and is currently included in the third aim of the global Sustainable Development Goals (SDGs) (Mashrafi et al., 2024).

IoT technology deployed in healthcare has accelerated, transforming conventional clinical monitoring into efficient, continuous, and predictive care delivery. Three-hundred studies were reviewed and published between 2020 and 2025 in PubMed and Scopus. This established that ensemble ML methods such as Random Forest (RF) and XGBoost, when incorporated with IoT-generated physiological data, routinely achieved predictive accuracies between 85% and 95% across diverse healthcare applications (Zonayed et al., 2025). The review further affirmed that explainable AI frameworks were increasingly being adopted to ensure clinical trust and facilitate the deployment of IoT-integrated prediction systems in healthcare environments. ML based IoT remote patient monitoring (RPM) system that gathers physiological data in real time from wearable devices employed supervised learning to detect anomaly, and risk classification. Experiment on a synthesized dataset stimulating real-world vitals demonstrated that RF and long short-term memory (LSTM) model achieved 93% accuracy in predicting critical health conditions. This highlighted the visibility of IoT-based continuous monitoring as a platform for integrated predictive clinical decision support (Merugu et al., 2025).

The application of supervised ML to healthcare prediction tasks has been witnessed in maternal health risk classification and clinical informatics research at large. Pi et al. (2025) adopted six ML methods for predictive model: Multi-Layer Perceptron (MLP), Logistic Regression (LR), Decision Tree (DT), RF, Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM). The MLP model was utilized as a basic model for this investigation since it performed the best among them according to assessment metrics like accuracy and precision. The accuracy of the model and ability to generalize were enhanced by data preparation, early halting to lower the danger of overfitting, and synthetic minority oversampling technique (SMOTE) to address the disparity in class. MLP architecture featured three layers that were hidden, of 256, 128, and 64 neurons, respectively, employing the Rectified Linear Unit (ReLU) activation mechanism and a SoftMax output layer. Training was discontinued if there was no improvement in the validation loss after 300 epochs. The Adam optimizer was employed to train the model for up to 10,000 epochs, a batch size of 32, a learning rate of 0.001, and a loss function for cross-entropy.

Desuky et al. (2024) optimized an SVM classifier using the Mud Ring Algorithm (MRA) and compared its performance against RF and K-Nearest Neighbors on a maternal health risk dataset, finding that the MRA-optimized SVM delivered highly competitive results across accuracy, G-mean, F-measure, Matthew correlation

coefficient (MCC), and Kappa metrics. The study reinforced the consensus that RF was among the most reliable classifiers for this dataset, while also observing that further validation across diverse populations is crucial before clinical adoption. Mutually, these findings contextualized the present study within a good comparative landscape, verifying with RF model which uses IoT-collected physiological makers to classify maternal health risk (Zonayed et al., 2025). Ensemble learning has become a dominant paradigm in medical prediction tasks due to its inherent ability to reduce variance, control overfitting, and model non-linear clinical relationship that single algorithm cannot capture. Khadidos et al. (2024) confirmed this principle through Quad-Ensemble ML Framework (QEML-MHRC) which integrated bagging, boosting, stacking, and voting across multiple base classifiers for maternal health risk prediction. Using 19 training and test configuration, the framework outperformed individual classifier and illustrated that the combination of diverse ensemble strategies tackled class imbalance, and there was no linearity challenge concurrently. In a multi-category prognosis, the study of coronary care unit patient using XGBoost, SVM, Naive Bayes, and LR showed the critical relevance of F1-score, AUC, and MCC for imbalanced multi-class classification tasks as it noted that AUC provided robust class separation measurement across all thresholds, while F1-score balances the precision-recall trade-off that raw accuracy masks. Their study confirmed that XGBoost outperformed all benchmarks across these inclusive metrics, reinforcing that multi-metric evaluation provides a more clinically reliable model assessment than single-metric reporting (Wang et al., 2022). LR offers a baseline predictive model but ML algorithms like RF and XGBoost improve forecast accuracy and highlight major risk variables through prominent features. Using SHapley Additive exPlanations (SHAP) analysis, ML models, precisely gradient boosting frameworks, successfully detected fetal and maternal health risks and produced forecasts that were easy to comprehend (Innab et al., 2024). These results revealed how ML might support and help targeted maternal care and lower health risk in low-resource settings. The pervasive research on the limitations of maternal health risk prediction involved the overwhelming dependence on a single dataset, i.e., the 1,014 Ministry of Human Resource Development (MHRD) records collected by Ahmed and Kashem from rural Bangladesh. Their study in 2020 noted that out of 13 studies reviewed, 10 employed this single dataset; this raised huge concerns about how ecologically valid it was, and how applicable the published models were. The generalization of any model trained exclusively on the dataset is limited by its geographical exclusivity to rural Bangladesh, its small size, and its lack of stated inclusion and exclusion criteria on the data. While recognizing this restriction and advocating for multi-site as well as multi-country data gathering as a priority for future research, the current study added to this body of work.

Similarly, Malde et al. (2025) investigated the feasibility of predicting maternal health risks in lower-middle-income and resource-constrained settings using sparse clinical data and vital signs. Using the publicly available dataset of 1,014 pregnant women from rural Bangladesh, the study evaluated multiple ML algorithms using physiological variables including age, blood pressure, body temperature, heart rate, and blood glucose level. Their findings demonstrated that maternal health risks could be predicted with useful accuracy using a limited number of routinely available physiological features, highlighting the potential of ML-based risk prediction systems for resource-constrained healthcare settings. The study further emphasized the importance of appropriate sampling strategies and model comparison in improving predictive performance.

Despite the strong model-fit performance metrics, the gap between algorithm development and clinical deployment remains wide across the AI maternal literature. An ML approach published in Koçak et al., (2024) highlighted that model performance on irregular vital sign data from Low- and Middle-Income Countries (LMICs) could degrade significantly without huge preprocessing pipelines, standardized IoT data collection protocols, and regulatory amenities for model deployment. Similarly, Malde et al. (2025) demonstrated the potential of ML for maternal health risk prediction in lower-middle-income settings using sparse vital-sign data, while emphasizing the importance of developing approaches that are appropriate for resource-constrained environments. The results highlighted that systemic integration of AI tools within larger healthcare delivery and policy framework was essential for long-term improvements in maternal health, while technical brilliance in model creation was required.

The examined literature indicated a strong scientific basis that established the creation of IoT integrated and ML-based maternal health risk classification system. With AUC value above 0.97, and recall above 0.88, for high-risk cases, RF emerges as the best algorithm for the domain in the most complete tests. Since ensemble learning techniques could reflect non-linear feature interactions and reduce variation, they clearly outperformed single classifiers. Blood pressure and blood glucose are the key physiological predictors of maternal risk. Alongside, the literature identified persistent and chief gaps in dataset diversity, feature completeness, model interpretability, system-level deployment evidence, and validation rigor as those which collectively define the research agenda for the next-generation maternal health AI systems. The present study acknowledged this.

3. Methodology

The current study encompassed preprocessing techniques which were used to prepare the dataset for different algorithm models. A variety of exploratory data analysis techniques were deployed to create a thorough overview of the data, including the number of variables, minimum and maximum values for each factor, description, information, correlation, explanation using various visualization techniques, and lastly, the implementation of

various proposed model using ML algorithm. Each algorithmic stage was formally defined in Figure 1.

A retrospective and data-driven methodology was used in this study; the data was gathered via a publicly accessible maternal health dataset via an IoT-based risk monitoring system (Ahmed & Kashem, 2020). Many medical facilities, such as hospitals, maternal healthcare facilities, and community clinics provided the data. The IoT system continuously monitored the physiological indicators of pregnant women and recorded the information for later analysis. In order to enable automated assessment of maternal health risk, this study aimed to develop and evaluate ML models.

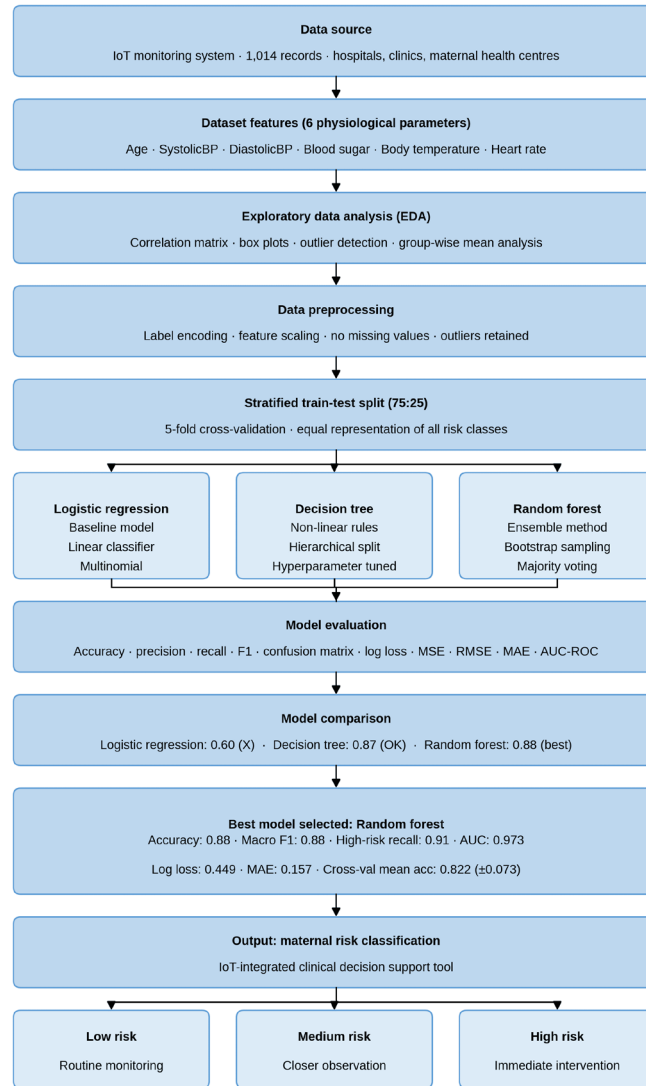


Figure 1. Methodology flowchart

3.1 Dataset Overview

Each of the records of the 1,014 pregnant women in the dataset was originally curated by Ahmed & Kashem (2020), covering 6 clinical and physiological parameters and 1 target variable. These included Age (years), Systolic Blood Pressure (SystolicBP, mmHg), Diastolic Blood Pressure (DiastolicBP, mmHg), Blood Glucose Level (BS, mmol/L), Body Temperature (BodyTemp, °F), and HeartRate (HeartRate, bpm). The target variable is the Risk Level, being divided into a three-class risk label.

3.2 Dataset Preprocessing

To guarantee data quality, consistency, and relevance for the creation of ML models, data preprocessing was conducted to categorize the data types and discover any missing values, thus contributing to a preliminary exploratory evaluation of the dataset. This study verified that every feature in the dataset was complete, with no missing entries.

A correlation matrix was employed to examine the pairwise linear relationships between all variables in the dataset simultaneously. It helped to detect where two predictors were too closely related, guided feature selection by identifying redundant or irrelevant variables, and validated the assumptions that dealt with statistical methods. Box plot visualizations were also used to discover outliers for each numerical feature, such as heart rate, body temperature, blood glucose level, systolic and diastolic blood pressure, and age. The evaluation of data distribution in relation to clinically acceptable ranges and the identification of extreme values were made possible by the box plots. Based on this study, the detected outliers were kept because they were clinically significant and indicated likely physiological abnormalities that could arise during high-risk pregnancies.

The dataset was categorized by risk levels, and the mean values of each characteristic were calculated in order to have a deeper understanding of the link between input features and target variable. This stage supported well-informed feature interpretation and model selection by revealing how physiological parameters varied among low, medium, and high-risk maternal groups.

After the exploratory research, multiclass classification was supported by encoding the target variables with numerical labels using Label Encoder. To enhance numerical stability and model convergence, feature scaling was employed. Lastly, stratified random sampling was employed for 75:25 split to divide the dataset into training and testing subsets, hence guaranteeing equal representation of all maternal risk classes in both sets, as formalized in Eq. (1), which defines the stratified train–test partition of dataset D .

Partition relations (D_{train} and D_{test} are disjoint and jointly exhaustive):

$$D = D_{\text{train}} \cup D_{\text{test}}, D_{\text{train}} \cap D_{\text{test}} = \emptyset \quad (1)$$

The partition sets D_{train} and D_{test} are disjoint and jointly exhaustive, as defined in Eq. (1). The dataset contains 1,014 samples, of which 760 (approximately 75%) are assigned to the training set and 254 (approximately 25%) to the test set.

Class-proportion preservation under stratification—for each risk class $C_k \in \{\text{Low, Mid, High}\}$, let $\pi_k = |C_k|/|D|$ be its proportion in the full dataset. Stratified sampling allocates instances to each subset in proportion to π_k :

$$|C_k \cap D_{\text{train}}| \approx \pi_k |D_{\text{train}}|, |C_k \cap D_{\text{test}}| \approx \pi_k |D_{\text{test}}|, \forall C_k \in \{\text{Low, Mid, High}\} \quad (2)$$

so that the within-subset class proportions both converge to the population proportion, up to integer-rounding error:

$$\pi_{k\text{-train}} = \frac{|C_k \cap D_{\text{train}}|}{|D_{\text{train}}|} \approx \pi_{k\text{-test}} = \frac{|C_k \cap D_{\text{test}}|}{|D_{\text{test}}|} \approx \pi_k \quad (3)$$

The dataset D was partitioned into disjoint training and test subsets, $D = D_{\text{train}} \cup D_{\text{test}}$, with $D_{\text{train}} \cap D_{\text{test}} = \emptyset$ (Eq. (1)). Of the 1,014 instances, 760 (approximately 75%) were assigned to D_{train} , while 254 (approximately 25%) were assigned to D_{test} . These proportions are approximate because the subset sizes must be integers. Stratification was applied so that each risk class $C_k \in \{\text{Low, Mid, High}\}$ retained approximately the same population proportion in both subsets (Eq. (2)), ensuring that the class distributions in D_{train} and D_{test} closely approximated that of the original dataset (Eq. (3)). The correlation matrix is displayed in Figure 2.

Towards evaluation of the linear relationship between continuous variables and identify possible multicollinearity, this matrix was created as a pre-modeling diagnostic step. Systolic-BP and Diastolic-BP had a strong correlation ($r = 0.79$), which implied that adding both at the same time to a predictive model could result in redundancy; this was taken into account when choosing features. For technical reasons, all other inter-variable correlations stayed below 0.5; this signified satisfactory independence. The matrix confirmed that the selected variables jointly covered a range of physiological features, supporting the combined use of subsequent investigation.

The boxplot as illustrated in Figure 3 was used during the preprocessing phase to access, spread, central tendency, and extreme values of each feature before model training. An important choice influenced by the analysis retained all detected outliers rather than remove them, because in a maternal health context, extreme physiological readings such as elevated blood sugar, very high blood pressure or abnormal body temperature were specifically the values that distinguished high risk pregnancies from low risk ones, and removing them would have resulted in a model that is biased against identifying the case it was built for due to data loss of clinically significant observation.

Towards obtaining a robust and unbiased estimate of the model's generalization performance, k -fold cross validation, where $k = 5$, was applied on the training instances. The training data, D_{train} , was partitioned into 5 mutually exclusive and equally sized folds. For each fold, $f \in \{1, 2, 3, 4, 5\}$, the model was trained on the remaining 4 folds and evaluated on fold f . The mean cross-validated accuracy is given by Eq. (4), the 5-fold cross validation accuracy, ($k = 5$).

$$CV_{cc} = (1/k) \sum f_1^k Acc_f, \quad (4)$$

$$Acc_f = (1/|D_f|) \sum_i^T (\hat{y}(X_i) = y_i)$$

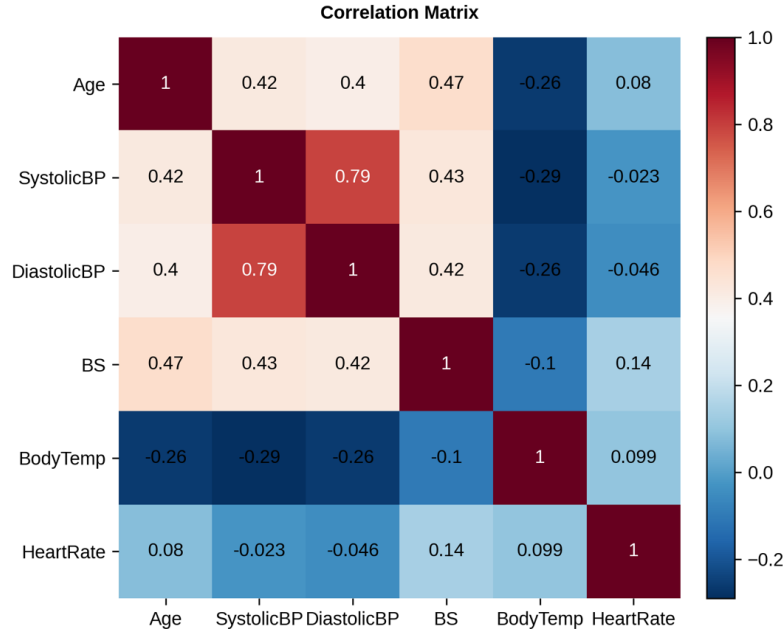


Figure 2. Diagram of correlation matrix
Note: BP = blood pressure; BS = blood glucose level.

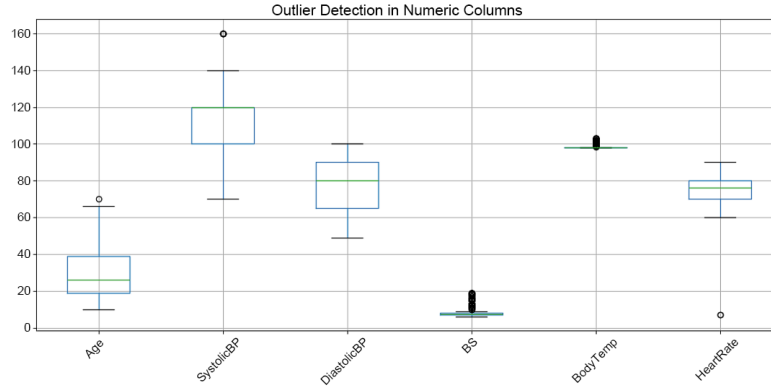


Figure 3. Box plot diagram
Note: BP = blood pressure; BS = blood glucose level.

3.3 Models Used

LR, DT, and RF, together forming the supervised learning models, were constructed and compared to determine the best method for predicting maternal health risks. These models were chosen because they could handle structured clinical data, and were widely employed in healthcare analytics.

LR: The baseline model used to evaluate the maternal risk groups' linear separability was LR. Multiclass classification was supported by a multinomial LR design to convergence and stability of numbers; feature scaling was used as well.

DT: Capturing non-linear correlations between risk levels and maternal physiological parameter led to the use of DT model. This method is interpretable and appropriate for clinical support-decision since it divides the feature space into hierarchical decisions. It lessens overfitting, and hyperparameters as the maximum tree depth are adjusted.

RF: This was used to increase prediction resilience and accuracy. It built several DTs, then adopted majority voting to aggregate their predictions. This method improved generalization and lowered variance, especially for complex and non-linear datasets. Based on comparative performance, this model became the chosen final prediction model.

3.4 Training and Assessment of Models

To ensure reproducibility of the experiment, a fixed random seed of 42 was used throughout the modelling procedures. The classifiers were applied using the following configurations:

LR: max_iter = 1000 and random_state = 42.

DT: random_state = 42, with other hyperparameters kept at their default values.

RF: n_estimators = 100 and random_state = 42, with the remaining hyperparameters retained at their default settings.

These settings were chosen to establish a reproducible baseline comparison rather than through an exhaustive hyperparameter-optimization procedure.

Feature scaling was performed after train-test split using Standard Scalar. The Standard Scalar was fitted exclusively on the training data, then used to transform both the training and held-out test data. This approach ensured that the test set was not used in estimating the scaling parameters, thus preventing data leakage. For model validation, a five-fold stratified cross-validation was applied exclusively to the training data using StratifiedKFold with shuffling enabled and random_state = 42. This cross-validation procedure was executed solely on the training data, while the held-out test set was reserved for the final and unbiased model evaluation.

Every model was assessed on a separate dataset test after being trained with the training subset. A wide range of assessment criteria were used to define both prediction, reliability, and classification accuracy; this measured the performance of the model. For every risk class, this measurement comprises macro and weighted-average scores, apart from accuracy, recall/precision, and F1-score. This is done to view how the classification performed, and spot misclassification trends among the low, medium, and high-risk categories, as well as the creation of correlation matrix. To evaluate calibration and prediction error, probability-based evaluation metrics as listed below were calculated: AUC, log loss, mean absolute error (MAE), root mean squared error (RMSE) and mean squared error (MSE).

3.5 Model Selection and Performance Comparison

Predictive performance and clinical relevance were taken into consideration when choosing a model. With an accuracy of 0.6, LR showed a limited ability of the non-linear correlations seen in the dataset. With an accuracy of 0.87, the DT model exhibited a significant improvement. RF model outperformed others with an accuracy of 0.88 and a balanced recall/precision in the three maternal risk categories.

With a recall of 0.91, RF model confirmed remarkable sensitivity to identifying high-risk pregnancies; this is crucial for early intervention and mother care. The capacity of the model to reduce false negatives (FNs) in the high-risk situations was further validated by the confusion matrix analysis. Thus, RF model was considered the best choice to predicting maternal health risks.

4. Results and Discussion

4.1 Descriptive Results

Each of the 1,014 entries in the maternal health dataset represented a pregnant woman under the observation of an IoT-based system. The target variable, i.e., the maternal risk level, was divided into Low Risk with a score of 0, Medium Risk (score 1), and High Risk (score 2). Systolic blood pressure, level of blood glucose, and diastolic blood pressure gradually increased from low- to high-risk in this category, in accordance with exploratory analysis using group-wise means, thus validating the clinical significance of the chosen features. High recall on high-risk class helps in situation where missed detections carry the greatest clinical cost.

4.2 Comparison of Model Performance

DT, LR, and RF were evaluated on the test dataset. Table 1 describes details of their overall classification accuracy.

Table 1. Summary of overall classification accuracy

Model	Accuracy
Logistic Regression (LR)	0.60
Decision Tree (DT)	0.87
Random Forest (RF)	0.88

LR exhibited the least satisfactory performance, thus suggesting that linear decision boundaries were insufficient for modelling the complex relationships among maternal physiological variables. The DT model

produced an improved performance, while RF derived the highest accuracy, and was therefore selected to be the final predictive model.

4.3 Confusion Matrix Analysis

RF correctly classified as follows:

- 67 out of 77 low-risk cases;
- 84 out of 98 medium-risk cases; and
- 72 out of 79 high-risk cases.

Low-risk patients had little misclassification though a few medium-risk and high-risk cases were mistaken for cases in nearby risk categories. Crucially, there were very few FNs in the high-risk group. This is important in clinical settings, where failing to detect high-risk pregnancy could cause repercussion.

4.4 Classification Metrics

The detailed classification performance of RF is presented in Table 2.

Within each class, the model did well and consistently. All recalls of 0.91 were attained by high-risk cases, demonstrating excellent sensitivity and efficient detection of fetuses in need of immediate medical intervention. The probability of incorrect classification of healthy pregnancies as high risk was decreased by the high precision (0.97) of low-risk and (0.91) of medium-risk classes.

Table 2. Classification report on Random Forest (RF) model

Risk Level	Precision	Recall	F1-score	Support
Low (0)	0.97	0.87	0.92	77
Medium (1)	0.91	0.86	0.88	98
High (2)	0.77	0.91	0.84	79
Macro average	0.89	0.88	0.88	254
Weighted average	0.89	0.88	0.88	254

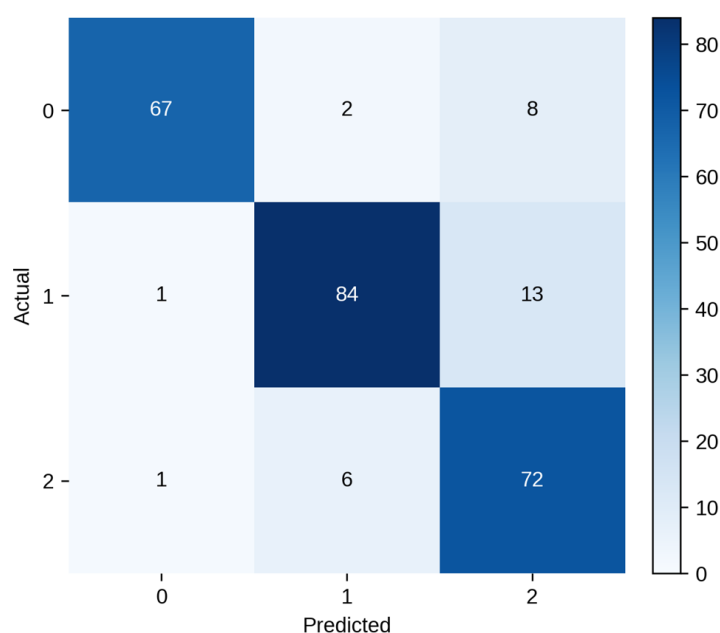


Figure 4. Confusion matrix

Figure 4 depicts the confusion matrix of the RF model achieved on the held-out test set. Rows represent the actual risk class, while that of the columns, the predicted risk class, (0 = Low, 1 = Medium, 2 = High). The model correctly classified 67 of 77 low-risk cases, 84 of 98 medium-risk cases, and 72 of 79 high-risk cases. Misclassifications were concentrated between adjacent risk categories (e.g., Medium predicted as High, or Low predicted as High) rather than between the extremes, and very few high-risk cases were missed which is consistent with the model's high-risk recall of 0.91 reported in Table 2.

4.5 Metrics for Error and Calibration

Probability-based error measures were calculated along with classification metrics; they were known as confusion matrix-derived metrics for assessing model calibration and prediction reliability. True positives (TPs), false positives (FPs), FNs, and true negatives (TNs) represent class k . Accuracy, recall, and F1-score were defined in Eqs. (5)–(7).

Eq. (5) describes classification accuracy, [Accuracy: 0.88].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Eq. (6) describes per-class recall, [Recall: 0.91].

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

Eq. (7) describes F1-score of accuracy, [F1-score: 0.88].

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

Supplementary error metrics, MSE (0.228), RMSE (0.478), and MAE (0.157), were calculated to assess the magnitude of prediction errors. These values suggested that misclassifications occurred mostly between adjacent risk categories (e.g., Low vs. Medium, or Medium vs. High) rather than between extremes (Low vs. High). This error pattern is clinically preferable, as it minimizes the risk of grossly misclassifying a high-risk pregnancy as a low-risk one. However, these metrics depend on the ordinal coding of risk levels (0, 1, 2), and assume equal clinical distance between categories, posing a limitation that should be acknowledged. The clinical relevance of the model was appropriately captured by the per-class recall and confusion matrix analysis, particularly the high recall of 0.91 for high-risk cases.

Log loss: 0.449—This affirms that the model does not just predict the correct risk class, but does so with realistic confidence, hence producing probability estimates that correlate with actual outcomes.

AUC: 0.973—This shows how well a model could distinguish between pregnancies at low, medium, and high risk across all decision thresholds. This result is consistent with top-performing models reported in literature on the same dataset, thus confirming RF approach as ideal for this classification problem. The final reported AUC was the macro average of the per-class AUC scores, treating all classes equally regardless of their sample sizes.

5-fold cross-validation mean accuracy: 0.822—This confirms that the robust performance of the model is not a coincidence due to promising data split. The result holds consistently across all five folds, indicating genuine generalization ability.

These findings demonstrated the applicability of RF model for predicting maternal health risk by showing that it generated well-calibrated probability estimates with comparatively low prediction error.

4.6 Limitations

The presence of duplicate records in the dataset is the limitation of this study. A meticulous check showed that 562 out of the 1,014 records were duplicates, representing 55.42% of the dataset. These records were retained as part of the publicly available dataset and were not removed during preprocessing. Though identical records were retained, they could potentially appear in both the training and test subsets during the random split. This was acknowledged as a limitation, as such repetition might contribute to optimistic estimates of the model's generalization performance. Hence, the findings should be interpreted with caution. Validation on independently collected datasets without repeated observations is ideal for future studies to ensure the robustness and generalizability of the proposed risk detection algorithm.

5. Conclusions

The study affirmed that routine physiological indicators gathered by IoT-based monitoring devices could categorize the risk level of maternal health using ML method. The results verified a well-trained ensemble model which could generate precise, dependable, and clinically significant risk estimates from a limited and regionally specific dataset across all evaluation criteria.

RF model performed well and evenly across all three maternal risk categories, with a macro F1-score of 0.88 and an overall accuracy of 0.88. The model's recall of 0.91 for high-risk pregnancies accurately identified almost all truly high-risk cases and reduced the risk of missed diagnoses. This is a vital requirement in maternal care,

where a missed high-risk classification could have life-threatening consequences, hence being the most clinically key finding. Given the natural overlap in physiological indicators across adjacent risk levels, the confusion matrix showed that misclassifications were mostly limited to neighboring risk categories, rather than representing extreme errors between low and high risks. This is a clinically acceptable and expected error pattern. The model's clinical credibility was further backed by probability-based measures. While a low MAE of 0.157 and RMSE of 0.478 suggested that prediction variations were small and well-defined, an AUC of 0.973 verified near-perfect performance that was biased in all risk categories. The model generated well-calibrated probability estimates, as shown by 0.449 log loss. This meant that medical practitioners could rely on both the projected label and the related confidence. The 5-fold cross validation mean accuracy of 0.822 provided added evidence that these findings were not the outcome of a favourable data split but true generalization ability instead.

Overall, the results established that ML, chiefly ensemble-based methods, pose a strong promise for automated maternal health risk detection and it offers a practical pathway toward early intervention, reduced diagnostic delays, and improved maternal outcomes in resource limited settings.

Author Contributions

Conceptualization, S.A. and O.K.A.; methodology, S.A.; validation, O.I.A.; investigation, S.A.; resources, O.I.A. and R.I.A.; data curation, S.A.; writing—original draft preparation, S.A., O.K.A., and A.K.A.; writing—review and editing, S.A., O.K.A., A.K.A., and R.I.A.; visualization, A.K.A. and O.I.A.; supervision, O.K.A. and A.K.A.; project administration, O.K.A. and A.K.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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